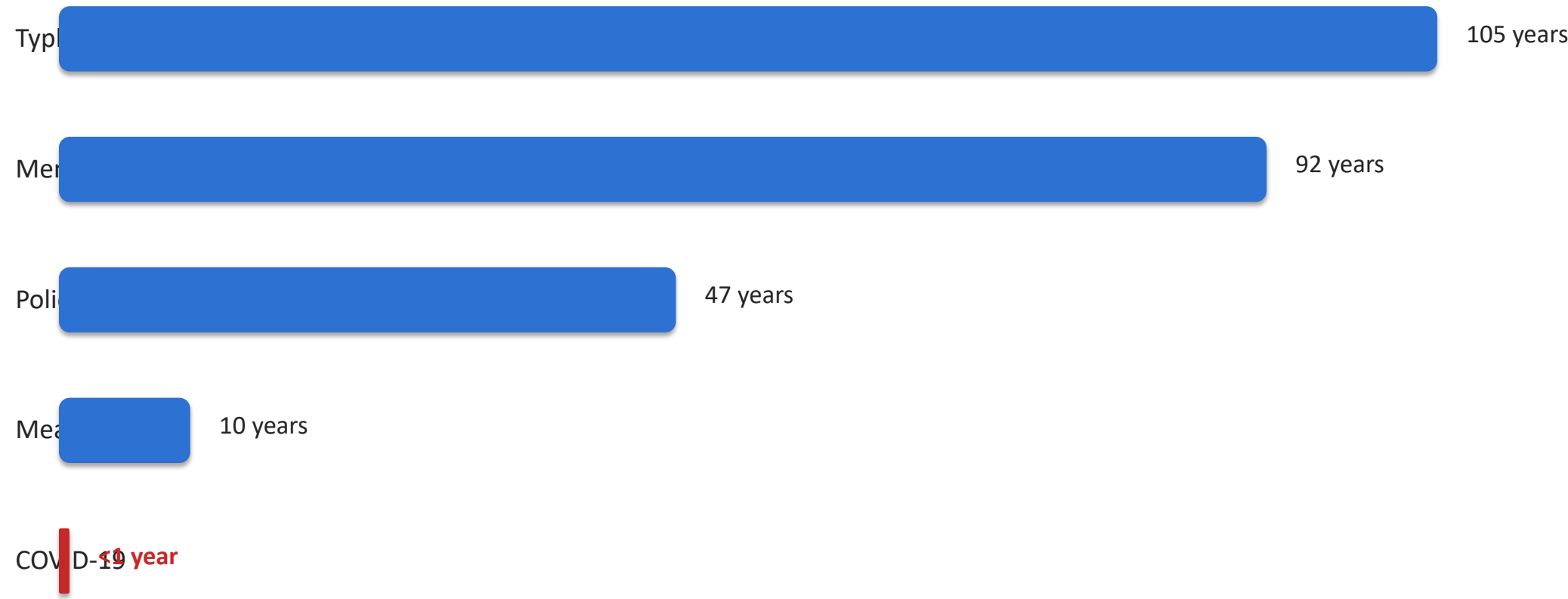


The Power of Vaccination: From Smallpox Eradication to C

A data-driven overview of accelerating vaccine innovation and measurable public-health impact (1796–2020)

Source: Our World in Data (ourworldindata.org/vaccination)

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

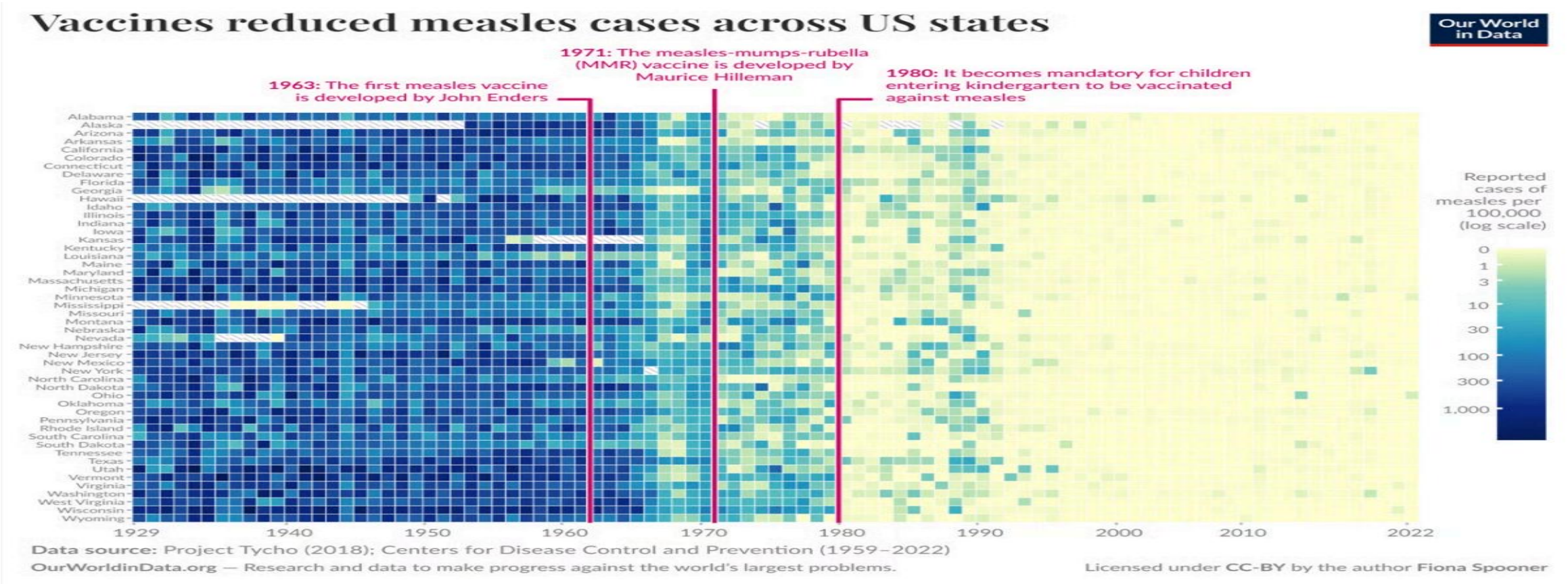


Pathogen identification → vaccine licensure gap shrank from 105 years (Typhoid) to under 1 year (COVID-19, 2020).

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-Vax Cases (per million/yr)	Post-Vax Cases (per million/yr)	Case Reduction	Pre-Vax Deaths (per million/yr)	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Measles	3,044	0.2	99.99%	2.5	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates



Download image or data

1963: First measles vaccine

1971: MMR vaccine

1980: Kindergarten mandate

Smallpox: The Only Human Disease Eradicated Through Vaccination



1959 WHO adopts global smallpox eradication resolution



1967 Intensified eradication campaign launched



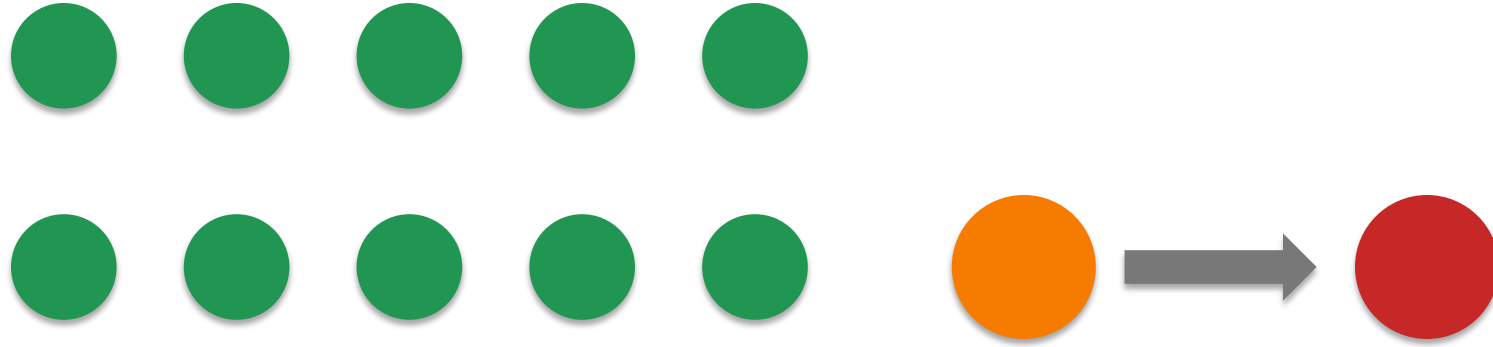
1977 Last natural smallpox case recorded (Somalia)



1980 WHO certifies global smallpox eradication

Why eradication succeeded: human-only reservoir, clearly visible symptoms, and an effective, low-cost vaccine.

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



Green = vaccinated individuals; Orange = vulnerable person (cannot be vaccinated); Red = infection source

High coverage interrupts transmission chains, indirectly protecting newborns, immunocompromised people, and patients on chemotherapy.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Pneumococcal case reduction

40.5%

Pneumococcal death reduction

31.3%

Hepatitis A case reduction

89%

Acute Hepatitis B case reduction

83.9%

- Access inequities limit routine immunization coverage
- Supply shortages interrupt delivery of doses
- Trust and scientific understanding still affect uptake

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- Vaccines achieved 100% elimination in the US for diphtheria, smallpox, and acute polio (cases and deaths).
- Development timelines collapsed from 105 years (typhoid) to under 1 year (COVID-19).
- Measles shifted from widespread high incidence to near zero across states after key policy milestones.
- Herd immunity extends protection to people who cannot be vaccinated.
- Remaining gaps (pneumococcal, hepatitis A, hepatitis B) require stronger access, coverage, and trust.

Call to Action: Invest in resilient global immunization systems to expand equitable access, sustain high coverage, and strengthen confidence in vaccines.